

2.4 DISINFLATION, NOMINAL AND REAL INTEREST RATES

Multiple Choice Questions

1990/AL/II/11

Which of the following statements about inflation are correct?

- (1) Inflation can exist in a barter economy.
- (2) In times of inflation, income will be redistributed from the general public to the government.
- (3) The nominal interest rate will be greater than the real interest rate when inflation is fully anticipated.
- (4) Inflation can be eliminated by price control.

- A. (1) and (3) only
B. (1) and (4) only
C. (2) and (3) only
D. (2) and (4) only

1990/AL/II/24

The inflation rate and the real interest rate were expected to be 5% and 4% respectively. In reality, the actual inflation rate turned out to be 7%. What was the real interest rate?

- [illegible]

1992/AL/II/21

If the nominal rate of interest is 10% but the real rate of interest is only 6%, the real return on holding cash will be

- A. 0%.
B. -4%.
C. -6%.
D. -10%.

1993/AL/II/29

If the sum of the real interest rate and the anticipated inflation rate is greater than the nominal interest rate, which of the following will be **FALSE**?

- A. Nobody likes to lend money to others.
- B. People want to borrow money for consumption.
- C. People want to make investments themselves.
- D. The anticipation inflation rate will fall.

1995/AL/II/19

If the inflation rate falls and remains positive,

- A. the economy is experiencing unemployment.
- B. the price level is falling at a fixed rate.
- C. the price level increases and then decreases.
- D. the price level is rising.

1996/AL/II/21

When inflation occurs, the opportunity cost of holding money is equal to

- A. the nominal rate of interest.
B. the expected rate of inflation.
C. the actual rate of inflation.
D. None of the above.

1999/AL/II/17

The inflation rate in an economy was 10% in 1996 and dropped to 5% in 1997. Which of the following must be true about the economy in the above period?

- A. There was a fall in the standard of living.
- B. There was a deflation in the year 1997.
- C. There was a rise in the prices of all goods.
- D. None of the above.

1999/AL/II/18

The expected inflation rate and the real rate of interest rate are 10% and 4% respectively. If the actual inflation rate turns out to be 9% only, the cost of holding cash is _____ and the real rate of return on holding cash is _____.

- A. 4% -5%
- B. 4% -9%
- C. 14% -9%
- D. 14% -10%

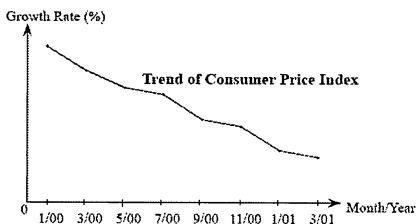
2002/AL/II/22

The nominal interest rate and the expected inflation rate are 10% and 7% respectively. If the actual inflation rate returns out to be 12%, then the realized real rate of interest will be _____ and the real rate of return on holding cash will be _____.

- A. -2% -12%
- B. -2% -10%
- C. 3% -12%
- D. 3% -10%

2002/AL/II/26

Study the diagram below.



The diagram shows that during the period,

- A. deflation occurred.
- B. the growth rate of money stock was greater than the growth rate of nominal output.
- C. the purchasing power of money was rising.
- D. the general price level was rising.

2003/AL/II/21

The real rate of return on holding non-interest-earning cash is positive if

- A. the rate of inflation is falling.
- B. the interest rate is falling.
- C. prices are falling.
- D. the velocity of circulation of money is falling.

2004/AL/II/15

Disinflation means that

- A. the inflation rate is at a disequilibrium level.
- B. inflation destabilizes the economy.
- C. the deflation rate is rising.
- D. the price level is increasing at a slower rate.

2004/AL/II/18

Suppose individuals can only choose to hold bonds or cash. The cost of holding cash is

- A. the real interest rate because the return from holding bonds will not be affected by a change in the price level.
- B. the real interest rate because it reflects the rate of return from investment.
- C. the nominal interest rate because it is the return from bonds.
- D. the nominal interest rate only when inflation rate can be accurately anticipated.

2005/AL/II/22

When there is deflation,

- A. the real interest rate will be negative.
- B. the nominal interest rate will be negative.
- C. the real interest rate will be higher than the nominal interest rate.
- D. All of the above.

2006/AL/II/08

If the nominal interest rate on one-year bond is 8% and the expected inflation rate is 3%, the nominal rate of return and the expected real rate of return of holding money are _____ and _____ respectively.

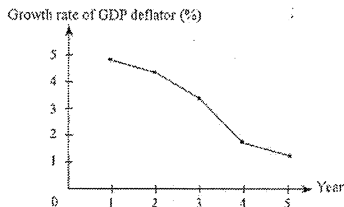
- | | |
|------------------|------------------|
| A. 0% -3% | B. 0% -5% |
| C. -8% -3% | D. -8% -5% |

2006/AL/II/21

The real interest rate will be higher than the nominal interest rate if

- A. the actual inflation rate is lower than the expected inflation rate.
- B. the expected inflation rate is falling.
- C. the nominal interest rate is falling.
- D. the price level is expected to fall.

2006/AL/II/23



The above diagram shows that during the period

- A. there is disinflation in the economy.
- B. the growth rate of money stock is positive.
- C. the general price level is falling.
- D. the standard of living is falling.

2008/AL/II/21

The nominal interest rate is the same as

- A. the real interest rate when people expect no change in the general price level.
- B. the real interest rate when the central bank holds the interest rate constant at a target level.
- C. the real return of holding monetary assets.
- D. the real interest rate plus the actual inflation rate.

2008/AL/II/23

When there is inflation,

- A. net monetary debtors will gain because the real value of interest payments will fall.
- B. the cost of holding money is the real interest rate instead of the nominal interest rate.
- C. a positive growth rate of real income implies a positive growth rate of nominal income.
- D. a positive nominal interest rate necessarily implies a positive real interest rate.

2010/AL/II/19

Suppose the nominal interest rate on a one-year bond is 3% and the expected deflation rate is 2%. However, the general price level turns out to rise by 3%. The nominal rate of return and the actual real rate of return of holding the bond are _____ and _____ respectively.

- A. 3% 5%
- B. 3% 0%
- C. 0% 5%
- D. 0% 0%

2011/AL/II/18

Suppose the market rate of interest is 5% and the expected inflation rate is 3%. The opportunity cost of holding money is _____, and the real rate of return on holding money is _____.

- A. -5% 2%
- B. 2% -3%
- C. 5% 3%
- D. 5% -3%

2012/AL/II/21

The nominal interest rate and the expected inflation rate are 8% and 5% respectively. If the actual inflation rate is 10%, the real rate of return on holding cash will be _____.

- A. -10%
- B. -8%
- C. -5%
- D. -3%

2014/DSE/I/32

The nominal interest rate of a one-year bond and the expected inflation rate are 8% and 3% respectively. If the actual inflation rate is 6%, then the actual real interest rate is _____ and the actual rate of return of holding cash is _____.

- A. 2% -6%
- B. 2% -8%
- C. 5% -6%
- D. 5% -8%

2015/DSE/I/31

If the nominal interest rate is 5% and the expected inflation rate is 3%, the cost of holding money is _____, the nominal return of holding money is _____ and the expected real interest rate is _____.

- A. 5% -3% 2%
- B. 5% 0% 2%
- C. 8% -3% 3%
- D. 8% 0% 3%

2018/DSE/I/34

Assume that the real interest rate is 6% and the expected inflation rate is 3%. If the actual inflation rate is 5%, the cost of holding money will be _____.

- A. 4%
- B. 6%
- C. 9%
- D. 11%

2019/DSE/I/32

In times of inflation, suppose the nominal interest rate of a bond is higher than the expected inflation rate. If the actual inflation rate turns out to be higher than the expected inflation rate,

- (1) the actual inflation rate will be higher than the nominal interest rate of the bond.
- (2) the expected real rate of return of the bond will be greater than the actual real rate of return.
- (3) the actual real rate of return of the bond may be negative.
- (4) the real rate of return of holding cash will be higher than zero.

- A. (1) and (2) only
- B. (1) and (4) only
- C. (2) and (3) only
- D. (3) and (4) only

2019/DSE/I/37

Under unexpected deflation, which of the following persons will lose?

- (1) Ms Mok holds a time deposit of one million dollars in a deposit taking company.
- (2) Mr Lo receives interest from Silver Bond of which the rate of return is linked to the inflation, subject to a minimum return of 3%.
- (3) Mr Chan signs a two-year contract with an online TV company to subscribe to its service by a fixed monthly payment.

- A. (1) only
- B. (2) only
- C. (3) only
- D. (2) and (3) only

2020/DSE/I/34

When there is unexpected inflation, which of the following persons will gain?

- A. a landlord who adjusts his rental according to the price index
- B. a creditor who has loaned out a sum of money at a floating interest rate
- C. an employer who signs a fixed nominal wage contract with his employees
- D. a retired civil servant who receives a fixed monthly pension

2024/DSE/I/36

Nominal interest rate on bonds	6%
Expected inflation rate	4%

Based on the above information, the _____ of holding money is 6%, and the expected real rate of return of holding _____ is -4%.

- A. cost.....money
- B. cost..... bonds
- C. nominal rate of return.....money
- D. nominal rate of return bonds

2025/DSE/I/35

Suppose the nominal interest rate remains constant. When the actual inflation rate is lower than the expected inflation rate, the actual real interest rate will be higher than the _____.

- (1) expected real interest rate
- (2) nominal interest rate
- (3) actual inflation rate

- A. (1) only
- B. (1) and (2) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Short & Structured Questions

1994/AL/II/2

Explain why people reduce their cash holdings and savings deposit balances in banks when there is high inflation. (8 marks)

2003/AL/II/2

- (a) What is meant by deflation? Is it the same as a fall in the inflation rate (i.e., disinflation)? (2 marks)
- (b) What is the relation between the nominal and real interest rates when people expect deflation? Will the nominal rate be higher or lower than the real rate? (4 marks)

2007/AL/II/1(a)

Name one similarity and one difference between the terms in each of the following pairs: deflation; disinflation. (3 marks)

2012/DSE/II/11(a)

To relieve the effect of inflation on citizens, the government has issued inflation-linked bonds (ibonds) with interest paid to bondholders every six months. The interest rate is positively related to the actual inflation rate of the last half-year period.

- (c) What is the opportunity cost of holding cash instead of ibonds? (2 marks)
- (d) Explain how ibonds would relieve the effect of inflation on their holders. (2 marks)

2013/DSE/II/3

- (a) What is the relationship between nominal interest rate and real interest rate? (2 marks)
- (b) "When there is actual deflation, the nominal interest rate will be lower than the real interest rate." Do you agree? Explain. (3 marks)

2014/DSE/II/5(c)(ii)

Study the following information about an economy.

Year	Consumer price index (CPI)
2012	120
2013	126

Suppose people did not expect any change in the price level from 2012 to 2013. Explain whether each of the following persons would gain or lose when the change in the price level in the above table occurred.

- (ii) In 2012, Marlene invested in a project with an expected nominal rate of return of 12% per year. In 2013, she discovered that the actual nominal rate of return was 15% per year. (3 marks)

- (a) What is the opportunity cost of holding money? Explain. (2 marks)

2022/DSE/II/6b

- (b) 'If deflation is expected, the real interest rate will be higher than the nominal interest rate.' Why? (2 marks)

2023/DSE/II/10b

In 2022, the Hong Kong Financial Secretary announced the issuance of Silver Bonds for citizens aged 60 or above to subscribe. The nominal interest rate of the Silver Bonds is equal to the actual inflation rate in Hong Kong, subject to a minimum of 4%.

- (b) Suppose the expected inflation rate is 4.5% and there turns out to be an actual deflation of 1%.
- Calculate the expected and the actual real rates of return on the Silver Bonds. (2 marks)
 - Based on your answer in (b)(i), explain whether holders of the Silver Bonds would gain or lose. (2 marks)

2025/DSE/II/9(Cross Topic)

In the 2024-25 Budget, the Financial Secretary proposed to adjust the rate of the hotel accommodation tax (HAT) from 0% to 3% with effect from 1 January 2025. Hotels and guesthouses would be required to pay the HAT at the rate of 3% on accommodation charges.

- Explain whether the proposed HAT is a progressive tax, a proportional tax or a regressive tax. (3 marks)
- An economist comments, 'Both the indirect tax revenue and the direct tax revenue of the Government may be affected by the above proposed HAT.' Justify the economist's comment. (4 marks)

There are receptionists, bell boys, room attendants and managers working in hotels.

- What economic principle is illustrated by the above phenomenon? List **THREE** reasons why the application of this principle can increase labour productivity. (4 marks)
- The following is a job advertisement for room attendants in a hotel.

Room attendants Major duty: tidying up hotel rooms Working hours: 10am – 2pm Salary: \$85 per hour

From the viewpoint of the hotel owners, give **ONE** advantage and **ONE** disadvantage of the above wage payment method over piece rate. (2 marks)

- Many hotels in Hong Kong charge a higher room rate during public holidays than on other days. Explain whether the opportunity cost of staying in those hotels in Hong Kong is *necessarily* higher during public holidays than on other days. (3 marks)